CSS

DAY 9

Units:

CSS uses units to define sizes (like width, height, font-size, margin, padding, etc.).

There are two main types:

1. Absolute Units
2. Relative Units

Absolute Units

These have fixed sizes and do not depend on the screen or parent element.

Unit	Meaning	Example
px	Pixels (most common unit, fixed size)	font-size: 16px;
cm	Centimeters	width: 5cm;
mm	Millimeters	height: 50mm;
in	Inches (1in = 96px)	width: 2in;
pt	Points (1pt = 1/72 of 1in)	font-size: 12pt;
pc	Picas (1pc = 12pt)	font-size: 1pc;

Relative Units

These depend on parent element size or viewport size.

Unit	Meaning	Example
%	Percentage of parent's value	width: 50%;
em	Relative to parent element's font-size	font-size: 2em;
rem	Relative to root (html) font-size	font-size: 1.5rem;
vw	% of viewport width	width: 50vw;
vh	% of viewport height	height: 100vh;
vmin	% of smaller dimension (width/height)	font-size: 5vmin;
vmax	% of larger dimension	font-size: 5vmax;
ch	Width of "0" character of current font	width: 30ch;
ex	Height of lowercase "x" (rarely used)	font-size: 2ex;

Absolute unit's examples:

```html
<body>
    <h1>css units</h1>
    <div>

    </div>
</body>
```

Px:

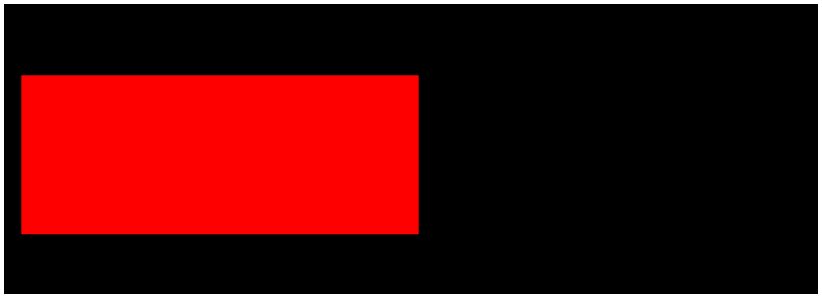
```css
div{
    width: 100px;
    border: 2px solid orange;
    background-color: yellow;
    height: 100px;
}
h1{
    font-size: 100px;
    color: white;
}
body{
    background-color: black;
}
```

```
div{
    width: 1px;
    border: 2px solid orange;
    background-color: yellow;

}
h1{
    font-size: 100px;
    color: white;
}
body{
    background-color: black;
}
```

Cm

```
div{
    width:5cm;
    height: 2cm;
    background-color: red;
}
```



Mm

```
div{
    width:5mm;
    height: 2mm;
    background-color: red;
}
```

In

```
div{
    width:1in;
    height: 1in;
    background-color: red;
}
```

Pt

```
div{
    width:5pt;
    height: 5pt;
    background-color: red;
}
```

Pc

```
div{
    width:5pc;
    height: 5pc;
    background-color: red;
}
```

Relative units examples:

em

```html
<body>
    <div>
        <p>em unit</p>
    </div>
</body>
```

em unit

```css
p{
    font-size: 2em;
}
body{
    font-size: 16px;
}
```

em unit

rem

```css
p{
    font-size: 2rem;
}
html{
    font-size: 16px;
}
```

em unit

%

```
div{
    width: 50%;
    background-color: green;
}
```

em unit

vh

```
body{
    border: 2px solid red;
    margin: 0;
    height: 30vh;
}
```

em unit

Lorem

Lorem

Lorem

Lorem

Lorem

vw

C:\Users\valib\OneDrive\Desktop\CSS60\units\style.css

```
body{
    border: 2px solid red;
    margin: 0;
    width: 50vw;
}
```

em unit

Lorem

Lorem

Lorem

Lorem

Lorem

em unit

Lorem

Lorem

Lorem

Lorem

Lorem

CSS Layouts:

- A layout in CSS means arranging elements on a webpage in a structured way.
- It defines how elements are placed, aligned, and sized within a container or page.
- Example: a navigation bar at the top, sidebar on the left, content in the centre, and footer at the bottom.

Earlier methods:

- Float → used before modern layouts, but hard to manage.
- Positioning (absolute, relative) → used for precise placement, but not flexible.
- Table Layouts → old way, not responsive.

Now, we mainly use:

- Flexbox (1D Layouts)
- CSS Grid (2D Layouts)

Float property in CSS:

The float property in CSS is used to position an element to the left or right inside its container, allowing text and inline elements to wrap around it.

```html
<body>
    <div>
        <img src="../Workshop/assets/Bahubali.jpeg" alt="">
        <h1>Bahubali</h1>
        <p>Lorem ipsum dolor sit, amet consectetur adipisicing e
    </div>
</body>
```

```css
img{
    height: 200px;
    width: 200px;
}
```

Bahubali

Lorem ipsum dolor sit, amet consectetur adipisicing elit. Temporibus, maxime.

```css
img{
    height: 200px;
    width: 200px;
    float: left;
}
```

Bahubali

Lorem ipsum dolor sit, amet consectetur adipisicing elit. Temporibus, maxime.

```
img{
    height: 200px;
    width: 200px;
    float: right;
}
```

Bahubali

Lorem ipsum dolor sit, amet consectetur adipisicing elit. Temporibus, maxime.

Flexbox:

- Flexbox (Flexible Box Layout) is a one-dimensional layout model in CSS.
- It is designed to arrange items in a row or column, with control over alignment, spacing, and ordering.
- Unlike float, flexbox makes it much easier to build responsive layouts.

Flexbox Terminology

1. Flex Container → The parent element with display: flex.
2. Flex Items → The child elements inside the flex container.

Container Properties:

Property	Values	Description
display	flex, inline-flex	Enables flexbox (block or inline container).
flex-direction	row, row-reverse, column, column-reverse	Direction of flex items.
flex-wrap	nowrap (default), wrap, wrap-reverse	Allows items to wrap into multiple lines.
justify-content	flex-start, flex-end, center, space-between, space-around, space-evenly	Aligns items **horizontally**.
align-items	flex-start, flex-end, center, stretch, baseline	Aligns items **vertically**.
align-content	Same values as above	Aligns multiple rows (only works when flex-wrap is used).

Item Properties:

Property	Description
order	Controls order of items (default = 0).
flex-grow	Defines how much an item can grow relative to others.
flex-shrink	Defines how much an item can shrink relative to others.
flex-basis	Defines the initial size of an item (before growing/shrinking).
flex	Shorthand → flex: grow shrink basis.
align-self	Overrides align-items for individual items.

Examples:

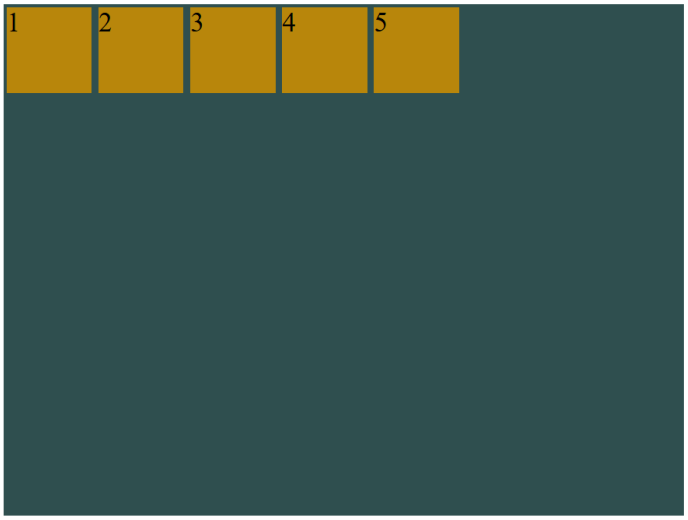
```html
<div class="parent">
    <div id="child1" class="child">1</div>
    <div id="child2" class="child">2</div>
    <div id="child3" class="child">3</div>
    <div id="child4" class="child">4</div>
    <div id="child5" class="child">5</div>
</div>
```

```css
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:auto;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

Container/Parent Properties:

display:flex

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:auto;
    display: flex;
}
```



flex-direction:

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

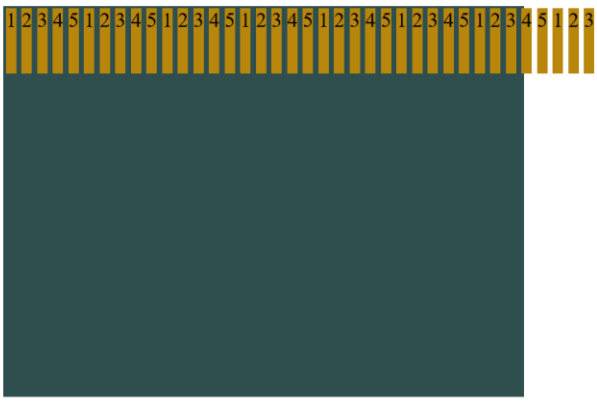
```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: column;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: column-reverse;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

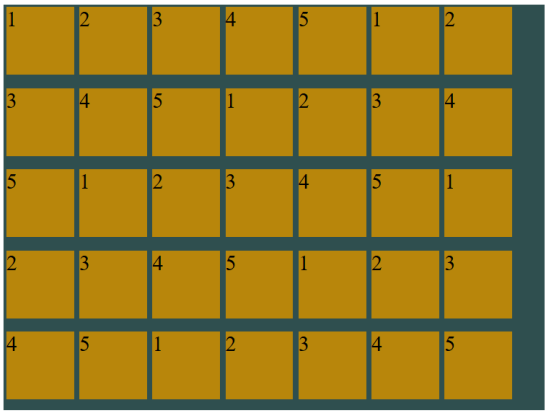
```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row-reverse;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

flex-wrap

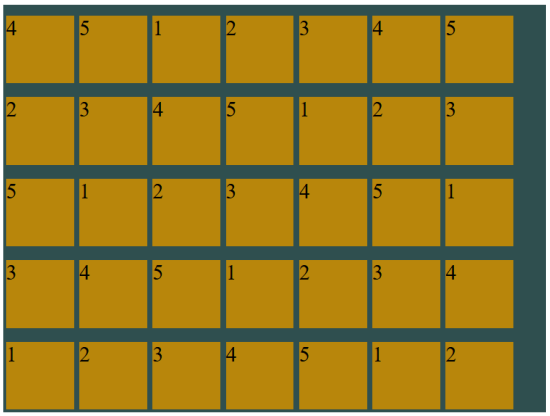
```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-wrap: nowrap;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```



```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-wrap: wrap;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

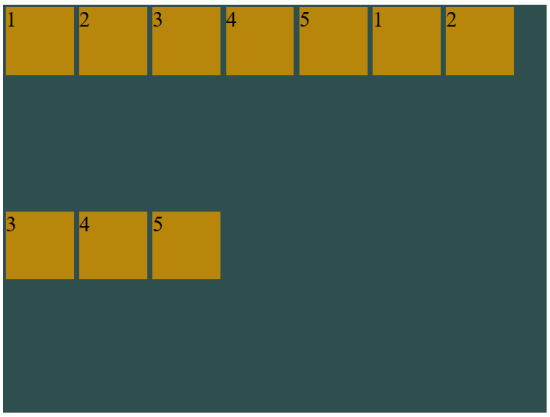



```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-wrap: wrap-reverse;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

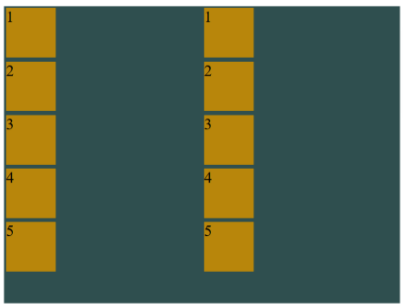


flex-flow:

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-flow: row wrap;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```




```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-flow: column wrap;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```



Justify-content:

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
    justify-content: center;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
    justify-content: start;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
    justify-content: end;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
    justify-content: space-between;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

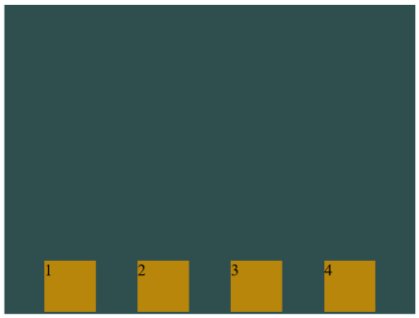
```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
    justify-content: space-evenly;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-direction: row;
    justify-content: space-around;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

align-items

```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    justify-content: space-evenly;
    align-items: center;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

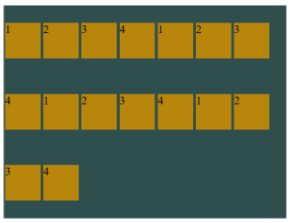
```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    justify-content: space-evenly;
    align-items: end;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```




```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    justify-content: space-evenly;
    align-items: start;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```

align-content

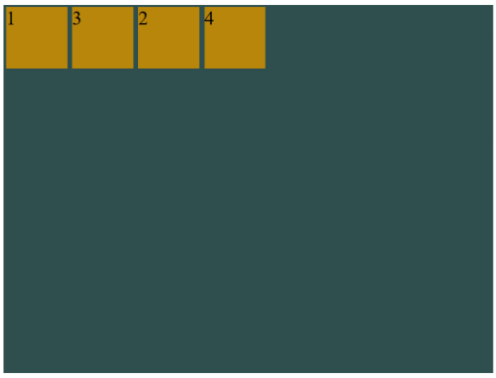
```
.parent{
    width: 400px;
    height: 300px;
    background-color: darkslategrey;
    margin:20px auto;
    display: flex;
    flex-wrap: wrap;
    align-content: space-around;
}
.child{
    width: 50px;
    height: 50px;
    background-color: darkgoldenrod;
    margin: 2px;
}
```



Item/child Properties:

Order

```
#child1{
    order: 1;
}
#child2{
    order: 4;
}
#child3{
    order: 3;
}
#child4{
    order: 10;
}
```



flex-grow

```
#child1{
    order: 0;
    flex-grow: 1;
}
#child2{
    order: 0;
}
#child3{
    order: 0;
    flex-grow: 1;
}
#child4{
    order: 0;
}
```

flex-shrink

```
#child1{
    flex-grow: 0;
    flex-shrink: 1;
}
#child2{
    order: 0;
    flex-shrink: 10;
}
#child3{
    order: 0;
    flex-grow: 0;
    flex-shrink: 0;
}
#child4{
    order: 0;
    flex-shrink: 10;
}
```

flex-basis

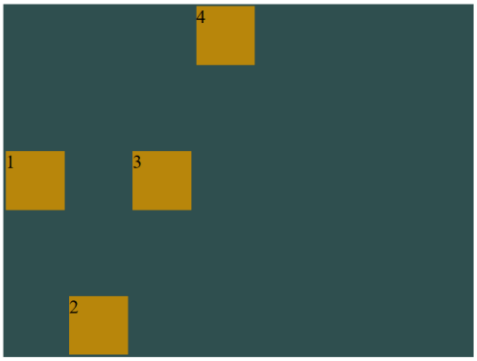
```
#child1{
    order: 0;
    flex-grow: 0;
    flex-shrink: 0;
    flex-basis: content;
}
#child2{
    order: 0;
    flex-grow: 0;
    flex-shrink: 0;
}
#child3{
    order: 0;
    flex-grow: 0;
    flex-shrink: 0;
    flex-basis: content;
}
#child4{
    order: 0;
    flex-grow: 0;
    flex-shrink: 0;
}
```

flex

```
#child1{
    order: 0;
    flex: 200px 1 0;

}
#child2{
    order: 0;
    flex-grow: 1;
    flex-shrink: 1;
}
#child3{
    order: 0;
    flex-grow: 1;
    flex-shrink: 1;

}
#child4{
    order: 0;
    flex-grow: 1;
    flex-shrink: 1;
}
```

align-self

```
#child1{

    align-self: center;

}
#child2{

    align-self: flex-end;
}
#child3{

    align-self: center;

}
#child4{

    align-self: flex-start;
}
```